Non-separability of a single industrial cluster from its regional background in satellite NO_2: a null result over Tangshan, China

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31 August 2026

Abstract

Tropospheric NO_2 from Sentinel-5P/TROPOMI is an established proxy for combustion activity and is increasingly proposed as a real-time economic nowcast. We test a deliberately harder, single-cluster case: whether a strictly public, free-data analysis can isolate the activity signal of *one* industrial cluster — the Tangshan blast-furnace (BF) complex in Hebei, China (39 located facilities, 26 active) — and validate it against an independent physical benchmark, the BF operating rate published by the Centre for Research on Energy and Clean Air (CREA). Five analyses, each targeting a different route by which the signal might be recovered, were computed blind to the benchmark at parameters fixed beforehand and judged under autocorrelation-robust inference. All five are negative. We report the outcome as an *exclusion* rather than a failure to reject: at an effective sample size of ≈ 289 weeks, these data rule out any weekly linear coupling above $r = 0.17$ (upper 95% bootstrap bound), i.e. under 3% of shared variance. A change-variance decomposition attributes 93–95% of the footprint's week-to-week variance to a regional common mode. A hard-to-falsify thermal proxy (public NASA active-fire data) tracks neither NO_2 nor the benchmark, because it observes agricultural crop-residue burning rather than furnaces. The encouraging 2023 peak in a rolling correlation reaches only $r = 0.276$ and is unremarkable under a rolling-window scan-maximum null ($p = 0.325$). A forecasting-skill screen finds no NO_2 feature that adds information over the operational benchmark. Finally NO_2 flux divergence (Beirle et al., 2021, 2023) returns a non-detection at the four real overpasses we could obtain. The near-null is not an artefact of the target: the CREA Tangshan rate tracks a separately surveyed 247-mill national panel at $r = 0.76$ in weekly changes, so measurement error in the benchmark cannot account for a feature correlation of 0.08. We are explicit that the five results are *not* statistically independent — they share one NO_2 cube, one background ring and one benchmark — so their convergence is corroboration only against the failure modes they do *not* share (Section 6.1), and we enumerate those. Two discriminating controls run against our own interpretation did not corroborate the spatial explanation: sweeping the background-ring radii to 150 km on a $4.8\times$ larger cube shows the cluster-local variance share is *ring-dependent* and still rising at the edge of the domain, and scoring the same features against national steel activity shows they track the national target no better than Tangshan's own. We therefore report spatial non-separability at TROPOMI's resolution as the leading hypothesis rather than an established cause, and state precisely what would settle it — a geostationary, higher-contrast NO_2 sensor (Kim et al., 2020; Park et al., 2025).

Keywords: tropospheric NO_2; TROPOMI; economic nowcasting; iron and steel; null results; equivalence bounds; source attribution; flux divergence.

Table 1: Overview of the five analyses. Each targets a different recovery route; all are computed blind to the validation benchmark and judged under autocorrelation-robust inference (Section 3). Three cross-cutting checks support all five rather than adding a sixth analysis: a spatial-scale sweep (Section 5.1), a background-ring radius sweep (Section 5.4) and a wider-scale benchmark contrast (Section 5.5). The five are not statistically independent: see Section 6.1.

Analysis	What it tests	Key statistic	Finding
Variance attribution	Cluster share of change-variance	regional 0.93–0.95; cluster-local ≤ 0.014 (95% bd.) at 25–60 km, 0.061 at 90–150 km	regional mode dominates; share is ring-dependent
Thermal proxy	Independent thermal signature of furnaces	$NO_2 \leftrightarrow$fire $r = -0.054$; CREA$\leftrightarrow$fire $r = -0.049$ (ns)	proxy uninformative
Regime stability	Robustness of the 2023 correlation peak	peak $r = 0.276$; scan-max $p = 0.325$	unremarkable under scan-max null
Forecasting skill	Lead/skill over the operational benchmark	$r = -0.014 \,/\, +0.063$; $\lvert r \rvert > 0.17$ excluded	no incremental info
Flux inversion	A separable point source above background	noise $\downarrow$ 3.4×; SNR $\approx +0.27$ at N=4	no source detected at N=4

1 Introduction

Premise. Satellite NO_2 is a combustion tracer, and a small literature uses it as an economic nowcast: at national scale for China (Morris and Zhang, 2019; Li and Zheng, 2023; Li et al., 2024), across global cities (Montgomery and Holloway, 2018), and as an industrial-shock read for Ukraine (Mao et al., 2025); the broad case for satellite data in economics is made by Donaldson and Storeygard (2016) and Ezran et al. (2023). We ask a harder, single-cluster question: can the NO_2 column over one industrial cluster — the Tangshan blast-furnace complex, the largest steel agglomeration in China (∼13% of national output) — track that cluster's physical output at a useful frequency, using only public, free data?

Why it is hard. Tangshan sits inside the Beijing–Tianjin–Hebei (BTH) plume, one of the most NO_2-loaded airsheds on Earth. Steel is only ∼30–43% of the cluster's own NO_2 column (Wen et al., 2024); traffic, power and residential sources fill the rest, and an ultra-low-emission retrofit programme is reported to drive a secular intensity decline that decouples emissions from output (Li et al., 2024). A validation is also only as good as its target, and the natural benchmark — the CREA BF operating rate (CREA, 2026) — is a commercial trader survey, so any near-null could be blamed on either the feature or the target. We therefore adopt a fixed analysis protocol (Section 3) and treat a well-characterised null as a valid, reportable outcome (Morris and Zhang, 2019).

Contribution and its limits. This paper reports five analyses, each targeting a different recovery route, and shows that they converge on one physical explanation. Table 1 is the map. We are deliberately careful about what that convergence buys: the five share their inputs, so they are not five independent draws on the same hypothesis. Section 6.1 sets out which failure modes are shared and which are not, and Section 7 lists, without hedging, the controls that would discriminate the remaining alternatives and that this study did not run.

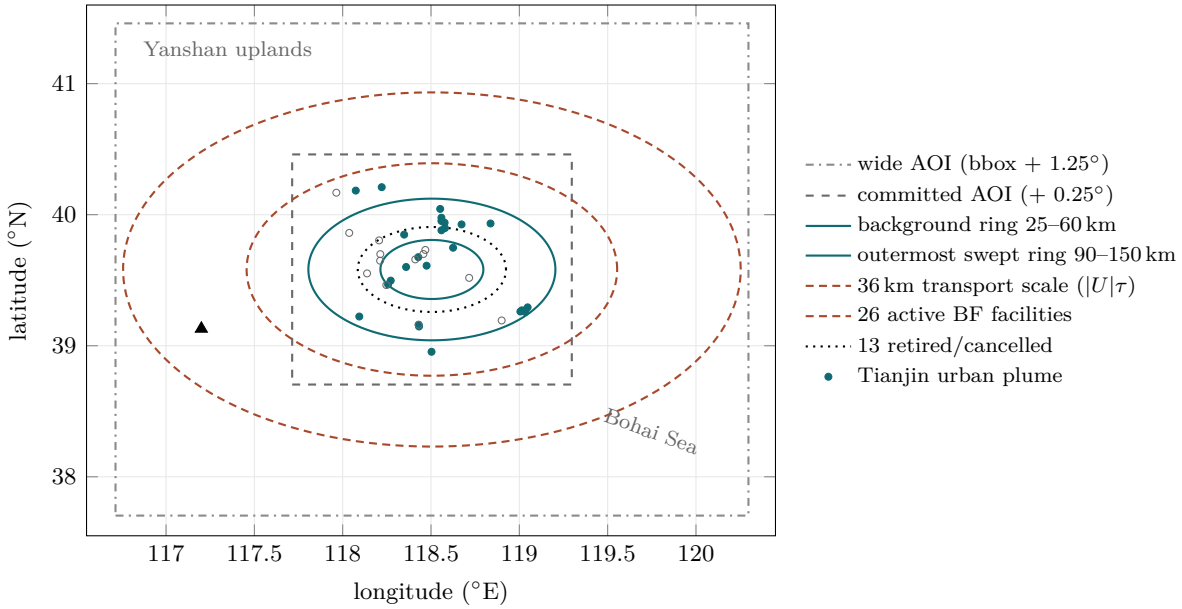


Figure 1: Study geometry, and the geometric confound the ring sweep runs into. The analysis footprint is the union of 15 km discs around the 26 active blast-furnace facilities (filled markers); the background reference of record is the 25–60 km ring (teal). The dotted circle is the mean advection scale implied by the flux analysis (mean effective wind $|U| = 2.48\,\mathrm{m\,s^{-1}}$ times the assumed NO_2 lifetime $\tau = 4$ h, i.e. ≈ 36 km): it falls *inside* the background ring, which is the self-subtraction concern of Section 5.3(i). The outermost ring the sweep could form (ember, 90–150 km; Section 5.4) escapes the advection scale but no longer samples the same air mass — it spans the Bohai Sea to the south-east and reaches the Tianjin urban plume to the west. That trade-off, not computation, is why the sweep stops at 150 km. Rings are drawn in ground distance about the cluster centroid.

2 Data and study region

Study region. The area of interest (AOI) encloses the 39 located Tangshan blast-furnace facilities (117.96–119.05 °E, 38.95–40.21 °N) plus a 0.25 ° buffer, on the North China Plain in Hebei (Fig. 1). Of the 39, 26 are active (25 *operating* plus one *operating pre-retirement*) and define the analysis footprint: the union of 15 km discs around the active facilities (448 of the 1,530 AOI grid cells). A concentric 25–60 km background ring around that footprint provides the regional-background reference used throughout. Analysis uses TROPOMI's native retrieval grid (∼4–6 km). The choice of ring radii is the single most consequential free geometric parameter in this study; it is examined critically in Section 5.3 and swept in Section 5.4, which required re-acquiring the record over a 1.25° buffer because the AOI defined here is too small to form a complete annulus beyond 63 km.

NO_2 **data and processor versions.** Tropospheric NO_2 was acquired from the Copernicus Data Space Ecosystem (CDSE) `SENTINEL_5P_L2` collection through openEO, server-side clipped to the AOI, and screened at `qa_value` ≥ 0.75. The archive comprises 2,641 AOI-intersecting overpasses from 2019-01-01 to 2026-06-12, composited without interpolation to a weekly cube (389 weeks) and a daily cube. Individual Level-2 orbit swaths, obtained separately from the CDSE OData/zipper route, supply per-overpass fields for the flux analysis.

We state plainly a limitation a reviewer will otherwise find: **the record is not processor-version homogenised.** The openEO collection does not expose a per-overpass processor version, so our manifest labels overpasses by a fixed 2022-07-01 cutoff (1,219 labelled `v1.x`, 1,422 `v2.x`). The record therefore spans the v1.2, v1.3, v1.4 and v2.x processor sequence, and no correction for the known step change in tropospheric columns at the v2 transition (and the associated PAL

Table 2: Reconciliation of every sample size in this paper. Each row is the count after applying the stated screen to the row above it within its block. No observation is imputed or interpolated anywhere in the pipeline.

Block	Screen applied	n
Daily cube (Section 5.2)		
	calendar days spanned, 2019-01-01 to 2026-06-12	2,720
	days with a valid footprint *and* ring value after cloud/QA screening	2,007
	↪ minus days lacking ERA5 covariates (deseason · daily)	2,003
	of the 2,007, those with a valid *immediately preceding* day	1,559
	↪ minus days lacking ERA5 covariates (first-diff · daily)	1,555
Weekly cube		
	weekly composite periods, 2019-01 to 2026-06	389
	↪ minus one period lost to differencing (first-diff · weekly)	388
	↪ weeks jointly observed with the CREA benchmark	379
	↪ minus weeks below the 0.25 valid-coverage screen	364
	↪ weeks jointly observed with the active-fire series (ends 2026-05-03)	373

reprocessing) was applied. Every result reported here is a *correlation or variance share* between two series measured on the same cube, which a common multiplicative level shift largely divides out; but any statement about a secular *trend* in the NO_2 level is confounded with this version sequence, and we withdraw such statements accordingly (Section 4).

Other data. (i) *Benchmark*: the CREA Tangshan BF operating rate (CREA, 2026), weekly, the only Tangshan-specific public physical-output series; CREA also publishes national aggregates (Section 7). (ii) *Meteorology*: ERA5 single-levels 10 m winds and boundary-layer height (Hersbach et al., 2020; C3S CDS, 2026), used as the ventilation covariate and, for the flux product, the advecting wind. (iii) *Thermal*: NASA active-fire detections (90,895 over 2019–2026) as a candidate physical proxy.

Sample sizes. Different analyses use different subsets, and the counts do not coincide. Table 2 reconciles every n quoted in this paper to its cause, so that no sample size appears without an accounted-for provenance.

The weekly cells of Table 4 use 389 (deseason) and 388 (first difference) because the variance decomposition never touches the benchmark and so is not subject to the CREA-overlap or coverage screens. Table 8 uses 364 for the background-corrected residual (which carries a `valid_coverage` column and is screened) and 379 for the intensity-decomposition covariate (which does not), and this is the whole of the difference between those two rows.

3 Methods

Estimation blind to the benchmark. Every feature, decomposition and screening statistic is computed from NO_2, meteorology and background alone; the CREA series is used only at the final scoring step. No parameter anywhere in the feature path is tuned to the benchmark. This guards against selecting a signal into existence.

The three NO_2 features, and how they differ. Three derived series appear in this paper, and because two of them are scored at the same n against the same benchmark we define all three here rather than let the reader infer them from context.

(i) **Background-corrected residual** (`no2_corrected`): the footprint column minus the concentric background-ring column, period by period. This is the raw spatial contrast, with no temporal model applied.

(ii) NO_2 **activity index** (`index_value`): the background-corrected residual after removing an annual harmonic, a heating-season structural term and a linear meteorological control (10 m winds, wind speed, boundary-layer height). This is the feature of record for the coupling tests.

(iii) **Intensity-decomposition covariate** (`residual_activity`): the activity component of the intensity decomposition. It is the *same series* as the activity index up to a fixed scale factor — their correlation is 0.999993 — and we say so plainly, because earlier drafts reported the two as though they were independent features.

It follows that the two headline weekly correlations against the benchmark, +0.078 and +0.063, are one feature under two screens, not two features: +0.078 ($n = 364$) applies the 0.25 valid-coverage screen, +0.063 ($n = 379$) does not. Only (i), at −0.014 ($n = 364$), is a genuinely different quantity. All three are reproduced by `analysis/benchmark_reliability.py` from the committed series.

Autocorrelation-robust inference. Both series are strongly serially correlated (for the weekly activity index against the benchmark, effective sample size $n_{\text{eff}} = 288.7$ of $n = 364$), so the naive Pearson p overstates significance. All coupling tests report, alongside the naive r: an effective sample size $n_{\text{eff}} = n(1 - ab)/(1 + ab)$ (with a, b the lag-1 autocorrelations) and a Newey–West heteroskedasticity- and autocorrelation-consistent variant (Newey and West, 1987); a moving-block bootstrap 95% confidence interval (CI) for r (Künsch, 1989); a block-permutation null p; and a Benjamini–Hochberg false-discovery-rate (FDR) correction across the test family (Benjamini and Hochberg, 1995). A single implementation of this inference is applied unchanged across all five analyses — which is convenient for consistency and is also, as Section 6.1 notes, a shared failure mode.

Reporting nulls as exclusions, not as failures to reject. A non-significant p is not a result; the quantity that is, is the largest effect the data can rule out. For every coupling test we therefore report the upper bound of the moving-block bootstrap CI on $|r|$ and the share of common variance it implies, alongside the minimum detectable correlation $1.96/\sqrt{n_{\text{eff}} - 3}$ that the design could have called significant. Across the three features of the preceding paragraph the widest such bound is 0.167 (activity index, screened; CI $[-0.024, +0.167]$, $n_{\text{eff}} = 288.7$, minimum detectable $|r| = 0.116$). The citable form of this paper's central null is therefore *an observed weekly coupling of* +0.078 *with any coupling above* $|r| = 0.17$ *excluded at 95% confidence — under 2.8% of shared variance* — and not "no significant relationship was found". Table 3 collects the bounds.

Fixed-parameter, benchmark-blind screens. Each analysis begins with a low-cost screen computed blind to the benchmark. Where a screen returns a null, the more elaborate model it would otherwise have justified is not pursued, and we report the screen result. Every parameter — window lengths, thresholds, block sizes, random seeds, 5,000 bootstrap and permutation draws — is a compiled-in default fixed in the codebase before the benchmark was read, so the decision to stop is determined by the data rather than by post-hoc choice.

This is not preregistration. There is no public registry entry, timestamp or protocol DOI for this study. Fixed seeds and compiled-in defaults are a weaker guarantee than a registered protocol: they establish that we did not tune parameters after seeing each outcome, but they cannot establish the order in which analyses were conceived. We use "fixed-parameter, benchmark-blind" throughout and avoid the word "preregistered", which we would not be able to support.

Table 3: Equivalence bounds for every NO_2 feature scored against the CREA benchmark, weekly. "excl. $|r|>$" is the larger absolute end of the moving-block bootstrap 95% CI — the smallest coupling the data are able to rule out — and r^2 is the share of common variance it corresponds to. "MDC" is the minimum detectable correlation $1.96/\sqrt{n_{\text{eff}}-3}$ at the autocorrelation-adjusted effective sample size, i.e. the smallest $|r|$ this design could have called significant. Features (i)–(iii) are defined in Section 3; rows two and four are the same series under two coverage screens. Reproduced by `analysis/benchmark_reliability.py`.

Feature	n	n_{eff}	r	boot 95% CI	excl. $\lvert r\rvert >$	MDC
Background-corrected residual (i)	364	295.1	−0.0138	[−0.120, +0.081]	0.120	0.115
NO_2 activity index (ii), screened	364	288.7	+0.0778	[−0.024, +0.167]	0.167	0.116
NO_2 activity index (ii), unscreened	379	291.3	+0.0624	[−0.049, +0.157]	0.157	0.115
Intensity-decomp. covariate (iii)	379	291.3	+0.0625	[−0.049, +0.157]	0.157	0.115

Sequential variance partition. The change-variance decomposition (Section 5.2) is a nested-OLS partition in a *fixed order*: regional → meteorological → cluster-local. Order matters. Variance shared between the regional term and the cluster-local term is attributed entirely to whichever enters first, which is the regional term by construction. The reported cluster-local share is therefore an estimate of the *incremental* explanatory power of a local term after the ring, not of the cluster's total contribution, and it is a lower bound on the latter. Section 5.3 explains why, for this geometry, that distinction is not cosmetic.

4 Baseline coupling

Before the decompositions, the plain result (Fig. 2). At the matched weekly cadence the deseasonalised, meteorology-controlled NO_2 activity index tracks the BF rate at $r = 0.078$ ($n = 364$, block-permutation $p = 0.21$, bootstrap 95% CI $[-0.024, +0.167]$); monthly, $r = 0.157$ ($n = 90$, $p = 0.24$, CI $[-0.079, +0.325]$). Neither is significant under any of the robust criteria. Stated as an exclusion rather than a non-rejection (Section 3): these data rule out any weekly linear coupling above $|r| = 0.17$, i.e. more than 2.8% of shared variance, at 95% confidence (Table 3).

On the "decoupling". The raw footprint NO_2 *level* correlates negatively with the BF rate over the whole record (monthly $r = -0.276$, $n = 90$), and an earlier partial record put this at -0.35. Splitting the record removes most of the sign: 2019–2021 $r = +0.168$ ($n = 36$), 2022–2026 $r = -0.089$ ($n = 54$). Subtracting the background ring removes it almost entirely (monthly $r = -0.032$). A whole-sample negative that disappears both within sub-periods and after background subtraction is what two slowly diverging series look like, not a within-period inverse relationship.

We previously read that divergence as physical — NO_2 falling with ultra-low-emission retrofits while output rose. **We withdraw that reading here.** As Section 2 sets out, the 2019–2026 NO_2 record spans an unharmonised TROPOMI processor-version sequence whose v2 transition is known to shift tropospheric columns by roughly 10–15%, in the same direction and on the same timescale. A secular trend in the level cannot be attributed to emissions rather than to retrieval versioning with these data. The correlation and variance-share results below are unaffected — they are computed from co-temporal contrasts between the footprint and its own ring, both drawn from the same retrieval — but the trend narrative is not supportable and we do not rely on it.

The one encouraging structure. A rolling correlation of the deseasonalised residual against the BF rate is regime-dependent (Fig. 2b). On the complete record, with the 78-week window and the deseasonalisation method of record, it ranges from -0.124 to $+0.276$, peaking on 2023-08-27. An exploratory earlier pass on a partial record, monthly and differently detrended, reported

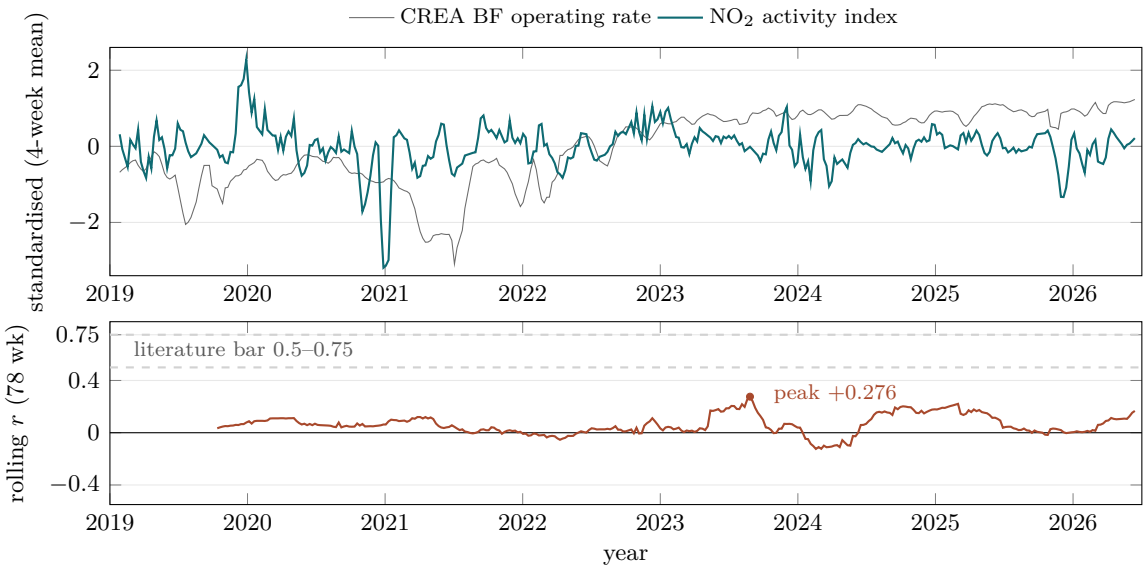


Figure 2: (a) The weekly NO_2 activity index and the CREA blast-furnace operating rate, both standardised and shown as 4-week means over the $n = 364$ jointly observed weeks. The benchmark's sharp curtailment troughs have no consistent counterpart in the NO_2 index; the lag-0 correlation is +0.078. (b) The 78-week rolling correlation of the deseasonalised residual against the benchmark over all 341 overlapping windows. It never reaches the 0.5–0.75 range reported in the nowcasting literature; its maximum is +0.276 on 2023-08-27, which Section 5.7 shows is unremarkable under a scan-maximum null.

a swing from −0.42 to +0.73 and thus appeared to cross the 0.5–0.75 range reported in the nowcasting literature; on the complete record with fixed parameters it does not come close. Section 5.7 tests whether even the +0.276 peak is more than favourable-window noise.

5 Results

5.1 No robust coupling at any spatial scale

A single-scale null invites the concern that a different AOI extent or grid resolution would rescue it (Parubets and Naito, 2025). We swept AOI extent (0.25° vs a strict 0.10° clip) × grid resolution (native, and block-averaged to ∼0.10°, ∼0.15° and ∼0.25°, by aggregation only, never interpolation), giving 8 scale cells. Each was evaluated at two cadences (weekly, monthly) × four detrending variants (level, year-on-year, STL, intensity-model), i.e. 64 lag-0 tests, each also re-evaluated at its selected peak lag over a 17-lag scan — 128 tests in all — and re-judged under the same robust + FDR standard.

No cell survives. Two of the 64 lag-0 tests are nominally significant on the naive Pearson p (the 0.10°/0.15° level cells: weekly $r = -0.132$, $p_{\text{naive}} = 0.011$, block-permutation $p = 0.034$; monthly $r = -0.229$, $p_{\text{naive}} = 0.030$, $p = 0.088$), and **neither survives the Benjamini–Hochberg correction across the family** ($p_{\text{adj}} = 0.996$ for both). None of the peak-lag selections survives it either. We report the two nominal hits rather than the summary "none is significant", which an earlier draft carried: a family of 128 tests producing two naive-p hits is exactly what the null predicts, and concealing them would misrepresent the sweep. The largest $|r|$ anywhere at lag 0 is 0.229. The sign of the levels relationship is itself unstable across scale, and the coarsest grid cannot resolve a source/background contrast at all. The tighter AOI does not rescue the signal ($\Delta|r| \approx 0.02$–0.05).

This is a cross-cutting robustness check on the AOI and grid parameters rather than a sixth analysis, and it strengthens the conclusion from "no signal at one scale" to "no robust coupling

Table 4: Change-variance decomposition (blind to the benchmark; all four cells reported). $\Delta R^2_{\text{local}}$ is the *raw* incremental R^2 of the cluster-local term entering last; the "cluster-local" share column clamps it at zero, since a negative incremental R^2 is not an admissible variance share. $\sqrt{f}$ is the implied change-detector correlation ceiling computed from the clamped share.

Basis × resolution	n	regional	meteo	$\Delta R^2_{\text{local}}$ (raw)	cluster-local	noise	ceiling $\sqrt{f}$
first-diff · weekly	388	0.944	0.005	−0.0209	0.000	0.051	0.00
deseason · weekly	389	0.951	0.002	+0.0064	0.006	0.040	0.08
first-diff · daily	1555	0.933	0.005	−0.0264	0.000	0.062	0.00
deseason · daily	2003	0.944	0.003	−0.0003	0.000	0.052	0.00

at any reasonable scale". Note what it does *not* sweep: the background ring radii, which are held at 25–60 km in every cell (Section 5.3).

5.2 The cluster-local change-variance share is ≈ 0

A change detector lives in the variance of *changes*. If the cluster-local component accounts for a fraction f of that change-variance and the remainder is independent noise, then $\sqrt{f}$ bounds the correlation attainable by a change detector built on it. We measured f directly, partitioning the variance of the footprint NO_2 changes (footprint minus the 25–60 km ring) into a regional common mode, a meteorological term, a cluster-local term and noise, across two change bases × two resolutions (Table 4, Fig. 3).

A ceiling argument we have removed. Earlier versions of this work stated that a *level* estimator is bounded by $\sqrt{0.40} \approx 0.63$, on the grounds that steel is ∼30–43% of the Tangshan column (Wen et al., 2024). That inference is invalid and we withdraw it. The 30–43% figure is a share of the column *mean*; $\sqrt{f}$ bounds a correlation only when f is a share of *variance*, and the two are unrelated in general — a component contributing 5% of a mean can contribute 90% of the variance around it, and a component contributing 40% of the mean can be perfectly constant and contribute none. The $\sqrt{f}$ ceiling reported in the last column of Table 4 is legitimate because there f *is* a measured variance share; the mean-share ceiling is not, and nothing in this paper now rests on it.

Three of the four cells return a small negative raw $\Delta R^2_{\text{local}}$ (-0.021, -0.026, -0.0003); the fourth returns $+0.006$. A negative incremental R^2 in a nested partition of this form means the local term buys nothing and its inclusion costs degrees of freedom — the cluster-local component is statistically indistinguishable from zero, and if anything the point estimate sits marginally on the wrong side of it. We report the raw values rather than only the clamped share because the clamping is otherwise invisible, and a table that shows a clean 0.000 without saying it was clamped misrepresents the estimate.

The measured cluster-local change-variance share is thus ≈ 0 *at this ring geometry*: the footprint and the 25–60 km ring move nearly in lockstep at both weekly and daily cadence. The hypothesis that a change detector has variance-share headroom over a level estimator is not supported at 25–60 km.

But the share is ring-dependent, and we now know it. The qualifier in the preceding sentence is not decorative. Section 5.4 reports a sensitivity sweep of the ring radii out to 90–150 km: the cluster-local share is *not* stable across ring geometries, and on one of the two change bases it rises by a factor of 8.7 as the ring moves outward — without plateauing — with a bootstrap confidence interval that excludes zero from 40–80 km onward. The ≈ 0 in Table 4 is therefore a property of the footprint-minus-25–60 km-ring contrast, not a scale-free property of

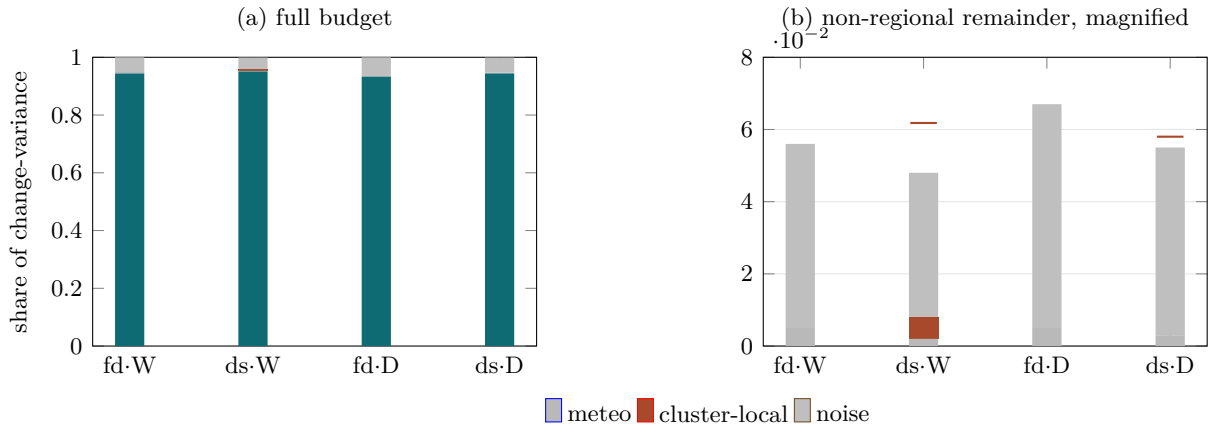


Figure 3: Change-variance budget per cell (W: weekly, D: daily; fd: first-difference, ds: deseasonalised). (a) The regional common mode fills the bar, which is the finding but also makes everything else unreadable. (b) The same bars with the regional term removed and the axis magnified 12.5×, so the terms that are actually in contention can be seen: the cluster-local slice (ember) is a hairline against noise, and in three of four cells its raw incremental R^2 is slightly *negative* (Table 4). Ember tick marks in (b) are the upper 95% moving-block bootstrap bounds on the cluster-local share — the quantity the null excludes rather than the point estimate it reports, at most 0.014 (weekly deseason) and indistinguishable from zero elsewhere. All values are at the 25–60 km ring; Section 5.4 shows they move with it.

the cluster, and we no longer present it as one.

5.3 Two ways this result could be an artefact of its own geometry

The result in Section 5.2 is the load-bearing one for this paper, so we state its two most serious alternative explanations explicitly. We were subsequently able to test the first and not the second; neither is excluded.

(i) The background ring may contain the cluster's own plume. The flux analysis (Section 5.9) measures a mean effective wind of $|U| = 2.48\,\mathrm{m\,s^{-1}}$ over the AOI and assumes a NO_2 lifetime $\tau = 4\,\mathrm{h}$. The implied advection length is $|U|\tau \approx 36\,\mathrm{km}$ — which lands near the middle of the 25–60 km background ring (Fig. 1). If a material part of the cluster's emitted NO_2 is present in the ring as advected plume, then subtracting the ring subtracts part of the signal along with the background, and the fixed regional → meteo → local partition order (Section 3) then books that shared variance to the "regional" term by construction. A cluster-local share of ≈ 0 is exactly what ring self-subtraction would also produce. We ran the discriminating sweep; Section 5.4 reports it.

(ii) Part of the "regional common mode" may be retrieval artefact rather than atmosphere. A 93–95% common mode between the footprint and its surrounding ring is a large number, and the TROPOMI retrieval supplies at least two mechanisms that would produce one without any atmospheric coherence. The stratosphere–troposphere separation subtracts a stratospheric field that is smoothed to roughly 1°, and the air-mass factor uses a priori NO_2 profiles from the TM5 chemistry-transport model at coarse resolution. Both fields are, by construction, spatially smooth across an AOI and a ring that together span well under 2°. Any residual error in either therefore appears as a common additive or multiplicative term shared almost perfectly between footprint and ring, and inflates the measured common-mode share mechanically.

We have not quantified this contribution, and it cannot be quantified from the level-3 composited product we use: separating it requires the per-overpass stratospheric column and AMF

Table 5: Cluster-local (and, in parentheses, regional) change-variance share by background-ring radius on the 1.25° cube, plain ring weighting. Only the ring varies; every other parameter is held at its Section 5.2 value. Every ring is a complete annulus (eight octants populated). The first-difference basis reports a clamped zero at *every* radius. Weekly $n = 389$ (deseason) / 388 (first-difference); daily $n = 2{,}001$ / 1,556.

Ring (km)	cells	deseason · W		deseason · D		first-diff (W & D)	
25–60	293	0.007	(0.954)	0.000	(0.946)	0.000	(0.947 / 0.935)
40–80	575	0.009	(0.936)	0.000	(0.942)	0.000	(0.927 / 0.931)
50–90	752	0.016	(0.911)	0.001	(0.926)	0.000	(0.903 / 0.914)
60–100	970	0.026	(0.884)	0.004	(0.907)	0.000	(0.884 / 0.896)
60–120	1,737	0.036	(0.854)	0.008	(0.883)	0.000	(0.863 / 0.872)
75–135	2,152	0.049	(0.805)	0.014	(0.839)	0.000	(0.821 / 0.831)
90–150	2,514	**0.061**	(0.758)	**0.023**	(0.793)	0.000	(0.781 / 0.788)

fields, propagated through the same compositing. The honest statement is that 0.93–0.95 is an upper bound on the *atmospheric* common mode, not an estimate of it. This does not overturn the paper's conclusion — a cluster signal that cannot be separated from a retrieval-borne common mode is just as unrecoverable in practice as one buried in a real regional plume — but it does change the physical interpretation, and a reviewer is entitled to that distinction.

5.4 Ring-radius sensitivity: the cluster-local share is ring-dependent

The concern in Section 5.3(i) is testable, so we tested it — twice. Holding the decomposition, the change bases, the cadences, the partition order, the covariates, the seed and the bootstrap depth all fixed, we swept only the background ring's radii. The decision rules were fixed before each run.

The first sweep was geometry-limited, and we say so. Run on the AOI of Section 2 (facility bbox + 0.25°), it reached 60–120 km and returned *inconclusive*. That AOI, however, extends only 63 km from the cluster centroid with full 360° coverage; past that radius its rectangle contributes four corner wedges, not an annulus, and the outermost ring the run could form was truncated at ≈ 124 km. Equal-octant weighting attenuates such anisotropy but cannot manufacture the missing azimuths. We therefore re-acquired the whole TROPOMI record over a 1.25° buffer — 4.8× the area, 2,741 overpasses, a 66×109 grid at the identical $0.055° \times 0.035°$ step — and re-ran the same sweep out to 90–150 km. All seven rings in Table 5 are complete annuli: all eight octants populated, realised outer radius within 0.03 km of the requested one.

The two cubes agree where they overlap. The wide grid's step matches but its origin is offset by a fraction of a cell, so the footprint discretises to 447 cells rather than 448 and the ERA5 footprint means shift by $\leq 0.05\,\mathrm{m\,s^{-1}}$. Rather than assume that immaterial, we re-ran the *committed* 25–60 km ring on the wide cube truncated to the original record's end. Every share agrees to within 0.0035 and every sample size to within 2 observations. The period-retention audit is likewise clean: the wide cube's retained weeks are a strict superset of the committed cube's (0 weeks lost, 11 gained, all in the tail added by the later download), because the period-coverage mask was evaluated over the original 0.25° sub-box precisely so that n could not move at the same time as the ring.

Widening the domain did not make the share converge. On the deseasonalised basis the cluster-local share rises monotonically with ring radius — weekly 0.007 → 0.061, a factor of 8.7; daily 0.000 → 0.023 — and its moving-block bootstrap interval excludes zero from the 40–80 km ring outward (weekly lower bound 0.001 → 0.023). The regional share falls from 0.95 to 0.76 over

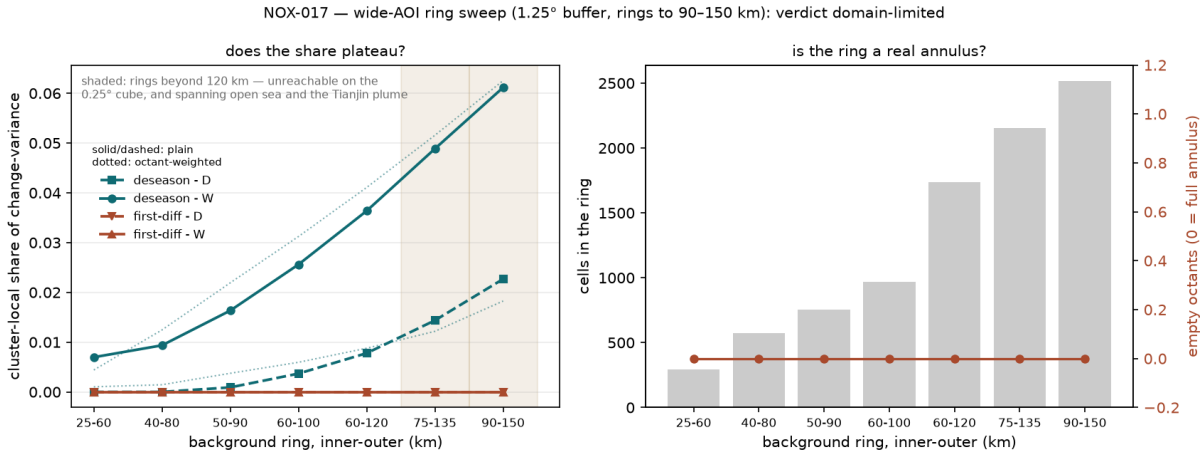


Figure 4: Left: cluster-local change-variance share against ring geometry on the 1.25° cube; dotted lines are the equal-octant-weighted variant. The shaded band marks the rings the 0.25° AOI could not form. The deseason basis rises monotonically and is still rising at the outermost ring; the first-difference basis does not move at all. Right: the geometry audit — ring size grows from 293 to 2,514 cells while the count of empty octants stays at zero, so every ring is a genuine annulus rather than a set of corner wedges.

the same sweep, and it falls on the first-difference basis too (0.947 → 0.781). At the outermost ring the weekly deseason cell clears, for the first time, both pre-registered materiality thresholds: its level (0.061) and its rise from the baseline ring (+0.054) each exceed 0.05.

Against that, three things. The quorum is not met — both rules require two of the four (basis × cadence) cells and only one fires. The first-difference basis is a clamped zero at *every* radius, so the variance leaving the regional term on that basis goes to meteorology and noise rather than to the cluster; a ring that were simply eating the cluster's plume should hand variance back to both bases once it moves away. And the share has *not* plateaued: the last increment is +0.012 weekly against a pre-registered plateau tolerance of 0.01. The sweep's own rule therefore returns **domain-limited**.

What we can and cannot now say. Self-subtraction is *not* established. Ring-robustness is not established either, and that is the change that matters: the ≈ 0 of Section 5.2 is demonstrably a property of the footprint-minus-25–60 km-ring contrast rather than of the cluster, and by 90–150 km the cluster-local share is an order of magnitude larger and no longer statistically indistinguishable from zero. The new information over the first sweep is negative and worth stating plainly: quadrupling the AOI area and pushing the ring to 150 km did not produce convergence. The ring radius is not a parameter that settles down over any domain this cluster's geography allows, so the share must be quoted as a range across ring geometries rather than at any single radius.

One limit is now unavoidable rather than incidental. At 90–150 km the background ring spans the Bohai Sea and the Tianjin urban plume: it is no longer the same air mass as the cluster. A share that keeps rising at that radius may be responding to a change of air mass rather than to escaping the cluster's own plume, and no sweep of this design can separate the two. Pushing the ring further would make that confound worse, not better — which is why we stop here and report the result as domain-limited rather than acquiring a wider cube again.

5.5 A wider-scale benchmark does not rescue the signal either

Section 7 of an earlier draft listed, as its second missing control, the cheap test that discriminates between the two explanations for the near-null: does the *same* feature track *wider-scale* steel

Table 6: Scale contrast, full record, benchmark-blind features scored unchanged against targets ordered by spatial scale. All four targets are shown; the appendix (Table 11) reports the complete family of 24 pairings including the two sub-period splits, over which the BH-FDR correction is computed. "Contrast" is the paired difference $r(\text{target}) - r(\text{Tangshan})$ with a moving-block bootstrap CI on the *common* sample — not two marginal intervals compared by eye; † marks a contrast whose CI excludes zero. *No pairing is robust-significant after FDR correction.*

Feature	Target (scale)	cad.	n	r	BH p	contrast vs Tangshan [95% CI]
intensity	Tangshan BF (cluster)	W	379	+0.062	0.466	— (baseline)
intensity	national BF rate	W	382	−0.094	0.366	$-0.165\ [-0.300, -0.018]^{\dagger}$
intensity	national crude steel	ME	89	−0.222	0.366	$-0.360\ [-0.647, +0.034]$
intensity	national pig iron	ME	89	−0.154	0.419	$-0.292\ [-0.532, +0.055]$
residual	Tangshan BF (cluster)	W	364	−0.014	0.864	— (baseline)
residual	national BF rate	W	367	−0.090	0.366	$-0.082\ [-0.233, +0.079]$
residual	national crude steel	ME	89	−0.120	0.523	$-0.103\ [-0.446, +0.379]$
residual	national pig iron	ME	89	−0.110	0.554	$-0.093\ [-0.376, +0.333]$

activity? If a Tangshan-AOI NO_2 feature tracks national steel better than it tracks Tangshan's own BF rate, the AOI column is carrying a regionally coherent component — the paper's spatial reading — and the feature is measuring something real about steel. We ran it (Table 6).

Widening the target does not strengthen the coupling. The national pairings are near-null too, and the paired contrast is negative throughout — the national targets are tracked, if anything, slightly worse than Tangshan's own. Under the pre-registered rule the outcome is *pipeline-suspect*: no national pairing reaches $|r| = 0.20$ on the primary target (best 0.094), far below the 0.5–0.75 range the national-scale nowcasting literature reports.

One contrast is worth stating rather than absorbing into the summary. For the intensity feature against the national BF rate the paired difference is $-0.165\ [-0.300, -0.018]$, and that interval *excludes zero*: the national target is tracked measurably *worse* than Tangshan's own, not merely no better. The direction is the opposite of what the spatial reading predicts — a Tangshan column dominated by a regionally coherent component should proxy the wider aggregate at least as well as the local one. We do not over-read a single contrast in a family of 24 whose FDR correction leaves nothing significant, but describing the family as uniformly null would omit its one directional result, and that result runs against our own interpretation.

The complete family, including the two sub-period splits, is in Table 11. We report all 24 rather than the eight full-record rows because three of the hidden cells reach nominal significance on the block-permutation test (intensity × national crude steel, 2019–2021, $r = -0.392$, $p = 0.036$; residual × national pig iron, 2022–2026, $r = +0.306$, $p = 0.037$; intensity × national pig iron, 2019–2021, $r = -0.365$, $p = 0.059$), and all three die under the family-wide correction. Showing only the rows that agree with the summary, while computing the correction over rows the reader cannot see, would be the same scan-selection error that Section 5.7 exists to diagnose.

Read this carefully in both directions. It does *not* establish that the pipeline is broken. Our feature is a Tangshan-AOI window — ∼13% of national steel, under 2° across — not national-coverage NO_2, so it was never guaranteed to proxy national output; the decisive form of that test needs a province- or nation-scale cube, which we did not build. Nor does it break the feature-versus-target ambiguity: all four targets are CREA/WIND survey products, so a survey-methodology artefact would move them together. The arm that would have added an independent publisher — NBS provincial and national monthly crude steel — was gated on a live feasibility probe, and the probe returned NO-GO: every NBS data endpoint refuses programmatic access, while static assets on the same host serve normally. Per our free-data rule we did not substitute a paywalled provider, so that gap stays open.

Table 7: Thermal triangulation, weekly, $n = 373$ (autocorrelation-robust). The fire signal tracks neither series, so it cannot arbitrate feature versus target.

Pairing	r (lag 0)	p (Newey–West)	p (perm)	BH p_{adj}	read
$NO_2 \leftrightarrow$ fire	-0.054	0.343	0.336	0.630	not significant
CREA $\leftrightarrow$ fire	-0.049	0.612	—	0.630	not significant

What it *does* establish is narrower and still consequential: the cheap test that would have corroborated the spatial explanation was run, and it did not corroborate it. We note also that the national correlations are strongly sub-period dependent (-0.34 to -0.39 over 2019–2021, collapsing toward zero over 2022–2026), which is inseparable from the unharmonised processor-version sequence of Section 2.

5.6 The public thermal proxy is uninformative here

To break the feature-versus-target ambiguity we sought a proxy that no statistical office can adjust: a space-based thermal signature over the facilities. The product tuned for sub-pixel high-temperature industrial sources (VIIRS Nightfire) is licence/pay-gated and thus outside a free-data scope; the free active-fire archive was therefore acquired in full and triangulated, blind to the benchmark, against both NO_2 and the CREA rate (Table 7).

The cause is physical: active-fire detections over this AOI are dominated by seasonal agricultural crop-residue burning — the detection climatology peaks in spring–autumn and is ≈ 0 in winter (normalised: Jan 0.00, May 0.87, Sep 1.00, Dec 0.09) with low radiative power ($\sim$1.4 MW median), a shape that persists even within 2 km of the facilities. Blast furnaces run year-round and are winter-*curtailed*, so a genuine furnace thermometer would peak in winter. The active-fire product does not detect the furnaces; the product that could (Nightfire) is behind a paywall. This is an informative proxy-validity result, not a claim that NO_2 is weak.

The benchmark is not the weak link, and this is measurable. The thermal route was meant to arbitrate feature against target and could not. A cheaper bound is available and we report it, because "perhaps the trader survey is simply noisy" would otherwise remain an unexamined escape from every result in this paper. CREA publishes, in the same release, a separately surveyed national panel — the blast-furnace starting rate across 247 mills. Over the study window the Tangshan rate tracks that panel at $r = +0.76$ in weekly *changes* ($n = 376$, moving-block bootstrap 95% CI $[+0.51, +0.85]$), and at $r = +0.26$ in levels. A series whose week-to-week movement is 76% correlated with an independently surveyed national panel is not dominated by survey noise. Under the usual attenuation argument its reliability is bounded below by that figure, so measurement error in the target caps the attainable correlation at $\sqrt{0.76} \approx 0.87$ — an order of magnitude above the 0.17 this study excludes (Table 3). The near-null cannot be explained by a noisy benchmark.

We are careful about what this does not do. Both series are published by CREA from WIND inputs, so this bounds the target's *reliability* — its reproducibility as a measurement — and not its *validity*: a shared survey-methodology artefact would move the Tangshan and national panels together and be invisible to this test. Closing that gap needs an independent publisher, which Section 5.5 reports was gated on a probe that returned NO-GO. What the test removes is the specific and otherwise untestable objection that the benchmark carries too little signal to correlate with anything.

5.7 The continuous coupling is near-null; the 2023 peak is a scan-maximum artefact

Exploratory analysis surfaced one encouraging pattern: a rolling correlation peaking in mid-2023. A rolling-window maximum is, however, a multiple-comparisons statistic. We tested it with fixed, benchmark-blind parameters and a scan-maximum null — a serial-dependence-preserving block permutation whose reference distribution is the maximum rolling-r over all 341 overlapping 78-week windows. The peak is $r = 0.276$ (2023-08-27) and sits at the **67.5th percentile** of its own scan-maximum null ($p = 0.325$): entirely unremarkable given how many windows were scanned.

Two correlations that must not be confused. Within the 78-week window that produced the peak, the intensity residual correlates with the benchmark at $r = 0.257$ ($n = 78$; the small difference from the 0.276 rolling value is the weekly re-alignment used by the jackknife routine). Over the *full record*, the same residual correlates with the same benchmark at $r = +0.063$ ($n = 379$; Table 8). These are not in conflict, and the 0.257 is not a full-record figure: it is the correlation inside the window that was selected *because* it maximised the correlation. Reporting it without that qualification — as an earlier draft of this paper did — would be the scan-maximum error the section exists to diagnose. The full-record value is the one that describes the coupling; the within-peak value describes the selection.

The jackknife rules out one alternative, not much more. A leave-one-event-out jackknife within the peak window removes the observations lying within ±4 weeks of each of the 21 sourced curtailment events, and re-measures. Removing the three highest-contribution events moves the within-window r from 0.257 to 0.349. We read this narrowly: no single curtailment event carries the peak, so the peak is not a one-event artefact. We do *not* read it as evidence of continuous coupling. Removing three events with a ±4-week exclusion drops roughly 27 of 78 observations, so the 0.349 is measured on $n \approx 51$ within an already scan-selected window; no significance is attached to it, and the change from 0.257 is well inside the sampling variability of a correlation on that many points. That the correlation *rises* when the largest curtailment events — the periods where a genuine activity→NO_2 response should be most visible — are removed is, if anything, evidence against a physical curtailment response and is not a result we can explain. The controlling number remains the full-record +0.063.

2024. The dual collapse diagnostic attributes the post-2023 fall-off to an intensity shift rather than a change in the curtailment calendar. Given Section 4, we no longer interpret that intensity shift physically: a slow change in the NO_2-per-unit-output ratio is not separable from processor-version drift in this record.

5.8 No forecasting skill over the operational benchmark

We next ask whether a public-data NO_2 nowcast can improve on the operational CREA benchmark. A screen measured, blind to the benchmark and at fixed parameters, whether any available NO_2 feature carries incremental information over the benchmark. The valid-proxy route is closed (Section 5.6), and neither NO_2 feature clears the bar (Table 8).

Both peak $|r|$ values fall below their per-lag white-noise bounds ($0.085 < 0.103$; $0.082 < 0.101$), before any correction for having scanned 17 lags. An earlier draft of this paper described these peaks as showing that "the feature follows the benchmark rather than leading it". That claim is withdrawn: it reads structure into two non-significant statistics, one of which is pinned to the boundary of the scan window, and it is precisely the error that Section 5.7 is devoted to identifying elsewhere. The correct statement is that no lead–lag structure is detectable at all.

Table 8: Forecasting-skill screen, weekly, blind to the benchmark. Both permutation p-values are far from significance and both bootstrap CIs straddle zero. The cross-correlation peaks are reported for completeness only: neither exceeds its own per-lag significance bound, and the first sits at -8 weeks, the edge of the scanned ± 8-week window, so it is a boundary selection and not an estimated lag. No lead or lag is identified by these data.

NO_2 feature	n	r (0)	p (perm)	boot 95% CI	peak lag / r	bound
Background-corrected residual	364	-0.014	0.815	$[-0.120, +0.081]$	-8 wk / -0.085	± 0.103
Intensity-decomp. covariate	379	$+0.063$	0.290	$[-0.049, +0.157]$	-2 wk / $+0.082$	± 0.101

Table 9: Market event study on NO_2-detected production events ($n = 27$ events per instrument): 5-day forward cumulative abnormal return relative to ACWI, with a bootstrap 95% CI, and the fraction of events with positive CAR. All four CIs contain zero and all hit rates are within sampling range of 0.5. The 108 event×instrument tests are *not* multiplicity-adjusted, so these are exploratory by construction. The upstream event detector is itself weak (4/47 production events matched; precision 0.15, recall 0.09).

Instrument	Exposure	CAR (5 d)	boot 95% CI	hit rate
BHP	diversified miner	$+0.0008$	$[-0.0136, +0.0120]$	0.48
RIO	diversified miner	$+0.0044$	$[-0.0074, +0.0140]$	0.41
VALE	iron-ore producer	-0.0020	$[-0.0227, +0.0157]$	0.48
SLX	steel-sector ETF	-0.0079	$[-0.0217, +0.0041]$	0.41

The screen is negative, so the fuller feature/baseline/surprise model it would have justified was not built. This is the expected consequence of Section 5.2: a residual dominated by the regional common mode cannot lead a cluster-activity benchmark.

A parallel market screen, reported in full. We also asked whether NO_2-derived event markers anticipate steel-sector equity moves. The detector flags 27 NO_2 events (16 surge, 11 drop) over 2019–2026; forward cumulative abnormal returns (CAR) were measured over a 5-day window against a global market index (ACWI), with the first tradeable session set at overpass +2 days so no look-ahead is possible (Table 9).

No tradable NO_2→market link is present in these data. We report the table rather than the one-line summary an earlier draft carried, because a claim of this kind with no n, no statistic and no method is not checkable and should not be made.

5.9 Flux divergence detects no separable source at the depth we could reach

The remaining route attacks the regional background directly: NO_2 *flux divergence* (Beirle et al., 2021, 2023). One forms the horizontal flux $\mathbf{F} = \Omega\,\mathbf{w}$ (column Ω times ERA5 wind $\mathbf{w}$) and takes its divergence $\nabla\cdot\mathbf{F}$, which isolates the local source term and should separate a Tangshan source from the advected background. Averaging many overpasses (each with its own wind, so advection dipoles cancel) should drive field noise down $\sim 1/\sqrt{N}$. The decisive question is whether the co-located source *separates* — SNR climbing toward the detection thresholds — or stays flat as the noise falls (Table 10, Fig. 5). The pipeline fixes two such thresholds in advance: SNR ≥ 3.0 for a clean detection (`snr_go`) and ≥ 1.5 for a partial-but-usable one (`snr_degraded`). Both are compiled-in constants, set before any real swath was processed. Figures below are referenced against the *lower* of the two, which is the more generous test.

What the numbers show. High NO_2 remains co-located with the plants (ratio 1.8–2.3×), but that is expected — the plants sit in the cluster's NO_2 core — and flux divergence does not convert co-location into a separable *source* here. The footprint source-term median (8.15 ×

Table 10: Multi-overpass scaling (real Level-2 swaths, blind to the benchmark). Noise is the median absolute deviation of the background divergence field ($\mathrm{mol\,m^{-2}\,s^{-1}}$). The last two columns are references, not fits: the $1/\sqrt{N}$ noise expectation from N=1, and the SNR a constant source term would produce if the noise fell as $1/\sqrt{N}$ from the N=1 value. There is one realisation per row, so no uncertainty is attached to any entry.

N	SNR	co-location ratio	noise (MAD)	$1/\sqrt{N}$ noise ref.	SNR if source constant
1	+0.27	2.06	3.618×10^{-8}	3.618×10^{-8}	+0.27
2	+0.22	2.29	1.770×10^{-8}	2.558×10^{-8}	+0.38
4	+0.27	1.80	1.068×10^{-8}	1.809×10^{-8}	+0.54

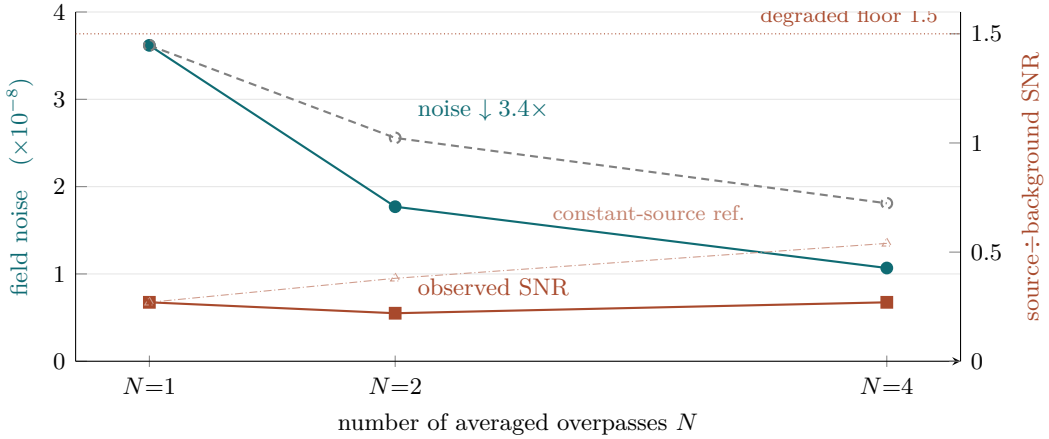


Figure 5: Signal-to-noise ratio versus averaging depth N over the four real overpasses obtained. Field noise (teal, left axis) falls 3.4× from N=1 to N=4, which is *faster* than the $1/\sqrt{N}$ reference (grey dashed) — see the text: this is not the confirmation of theory it may appear to be. The observed SNR (solid ember, right axis) stays near +0.27, well below both the 1.5 degraded-detection floor and the dash-dotted reference showing where the SNR would sit if a constant source term were divided by an ideal $1/\sqrt{N}$ noise. Three points, one realisation each, no error bars: the figure is a description, not a test.

$10^{-9}\,\mathrm{mol\,m^{-2}\,s^{-1}}$) sits near the background level (5.31×10^{-9}), both far below the N=4 noise (1.07×10^{-8}). The observed SNR at N=4 is 0.27 against the 0.54 that a constant source term over ideally averaging noise would give, so the numerator is falling roughly in step with the denominator.

Why N=4 does not license a strong claim. We reached four overpasses of an intended ten to twelve; the CDSE zipper endpoint throttled sustained pulls to $\sim$1.6 $\mathrm{MB\,s^{-1}}$, and the access API offers no server-side spatial subsetting, so a full-record inversion means $\sim$2.3 TB of transfer. Four is two to three orders of magnitude below the operating point of the method we are applying: the published TROPOMI point-source catalogues are built from two to three years of near-daily coverage, i.e. of order 10^2–10^3 usable overpasses per location (Beirle et al., 2021, 2023). The throttling explains why we stopped at four; it does not make four sufficient.

We had previously offered a scaling extrapolation — "if SNR grows as $\sqrt{N}$, lifting 0.27 to the floor needs $N \approx 124$ from N=4, or $N \approx 31$ from N=1" — as an acquisition estimate. It is withdrawn as such, for two reasons. First, the two figures are mutually inconsistent under a single model, which is the signature of an extrapolation whose premise does not hold: $\sqrt{N}$ growth from N=1 predicts SNR(4) = 0.54, and the measured value is 0.27. The data in Table 10 therefore *falsify* the premise of the extrapolation over the only range in which we can check it. Second, the target is misstated. The 1.5 threshold is the pipeline's `snr_degraded` level — "partial but usable" — whereas a clean detection is defined at `snr_go` = 3.0; reaching that from 0.27 under the same (falsified) $\sqrt{N}$ assumption would need $N \approx 494$. The defensible statement is conditional

and we leave it in that form: *if* the flat SNR over $N = 1$–4 is sampling noise around a genuine $\sqrt{N}$ growth, then order 10^2 overpasses would be required to reach even the degraded threshold; if it is not, no achievable N helps. Four overpasses cannot distinguish the two cases, which is precisely the point of this subsection.

An earlier draft argued that the *flatness* of the SNR trend, rather than the absolute N, carried the verdict. We weaken that argument to what it can support. Three points with one realisation each and no uncertainty estimate cannot distinguish a flat trend from a growing one: the spread among them (0.27, 0.22, 0.27) is of the same order as the growth $\sqrt{N}$ would produce over so short a range, and no trend test on $n = 3$ is meaningful. The co-location ratio is also non-monotonic ($2.06 \to 2.29 \to 1.80$) and is *lowest* at N=4, whereas oversampling that was revealing a persistent source should raise it. What the data do support is the weaker, comparative statement: at N=4 the SNR is 0.27 where a constant source over $1/\sqrt{N}$ noise would put it at 0.54, and it is nowhere near 1.5. That is a non-detection at the depth achieved, not a powered exclusion.

Why the noise falling faster than $1/\sqrt{N}$ is a warning, not a confirmation. The measured MAD falls to 1.068×10^{-8} where $1/\sqrt{N}$ predicts 1.809×10^{-8}: a factor 1.7 faster than independent pixel noise can average down. An earlier draft read this as the procedure "behaving as theory predicts". It is the opposite. For i.i.d. noise, $1/\sqrt{N}$ is the limit, not a floor to beat; beating it means the averaging is cancelling *coherent* structure, not noise. Candidate mechanisms include wind-dependent misregistration of the source term between overpasses (each overpass places the divergence signature in a slightly different place, so averaging smears and cancels it), and a changing set of QA-valid pixels over which the MAD is computed as coverage accumulates with N.

This matters because it makes the central observation ambiguous. "The numerator falls in step with the noise" is consistent with *there is no source* — but it is equally consistent with *the averaging procedure is cancelling the source along with the coherent background*. The four overpasses we have cannot distinguish these. Discriminating between them requires either a per-overpass wind-rotated composite (aligning each overpass's source signature before averaging) or a synthetic source of realistic amplitude injected into the real Level-2 fields and pushed through the identical pipeline. Neither was done.

What the synthetic control does and does not establish. The implementation carries a synthetic source-injection control: a Gaussian bump ($\sigma = 10\,\mathrm{km}$) is added to a simulated field and the same estimator is run. In that control the SNR does rise with N and the verdict flips to a detection, which verifies that the estimator is wired correctly and is not structurally incapable of finding a source. We are explicit that this is *all* it verifies. The control field is idealised in every respect that matters for power: pixels are uniformly distributed, there are no cloud gaps and no QA rejections, winds are set to a constant $3\,\mathrm{m\,s^{-1}}$ spread evenly around the compass so the advection dipole cancels exactly, and the injected source has amplitude 1.5×10^{-4} against a background column of 8×10^{-5} and a pixel noise of 1.3×10^{-5} — roughly eleven times the pixel noise and twice the background column. It also runs at N=8, twice the depth of the real analysis.

A positive control with a very large source in an idealised field establishes procedure correctness. It does not establish statistical power to detect a source of Tangshan's plausible magnitude in four real, cloud-gapped, irregularly-sampled overpasses. The claim that this is a "well-powered null", which an earlier draft made, is not supported by this control and is withdrawn. Establishing power requires injecting a source of realistic amplitude into the real Level-2 fields at the real N and measuring the recovery rate; that experiment is the designated next step and is listed in Section 7.

Status of this analysis. Flux divergence returns a non-detection at N=4. It is the weakest of the five results in isolation, and we present it as corroborating rather than load-bearing. Its value in this paper is chiefly that it is the one analysis that uses neither the composited cube, nor the background ring, nor the benchmark (Section 6.1).

6 Synthesis: one finding, seen from five angles

The five results were obtained by different methods — a variance partition, a thermal proxy, a rolling-regime test, a forecasting-skill screen, and a flux inversion — and they converge. Each analysis, in its own vocabulary, meets the same wall. Section 5.2 names it directly: 93–95% of the change-variance is a common mode shared with the surrounding ring. Section 5.7 is the same fact in the time domain: apparent coupling is a favourable window that regresses to the mean. Section 5.8 is its forecasting corollary; Section 5.9 its spatial confirmation; Section 5.6 forecloses the one external instrument that might have arbitrated, for a separate physical reason.

The spatial explanation, and why we now hold it more loosely. The reading these five results invite is a spatial one: Tangshan and its regional background are coherent at TROPOMI's resolution, the cluster column and the surrounding BTH column rise and fall together, and no amount of de-seasonalising, de-weathering, higher-frequency sampling or refined inference separates two signals that move together in space. On that reading additional TROPOMI overpasses would not help, because the source-versus-background *numerator*, not the noise, is the limiting quantity.

We are no longer able to present that as the explanation the evidence selects. Two controls designed to discriminate it from the alternatives have since been run, and neither corroborated it. The ring sweep (Section 5.4) shows the cluster-local share is a property of the particular background ring rather than a scale-free property of the cluster, and returns *domain-limited*: on a cube re-acquired over 4.8× the area, the share is still rising at 90–150 km, so whether the ≈ 0 survives at wider ring geometries is not settled by any domain this cluster's geography allows. The scale contrast (Section 5.5) shows the same feature tracks *national* steel activity no better than it tracks Tangshan's own rate — the opposite of what a regionally-dominated AOI column should do, and the single cheapest positive test the spatial reading had available to it.

Neither result overturns the near-null itself, which is measured independently of both the ring and the choice of target. What they remove is the confidence with which this paper can name its cause. The spatial explanation remains the most economical account of the five nulls; it is now a hypothesis this study failed to corroborate rather than a conclusion it established, and Section 8 says what would settle it.

6.1 What the convergence is and is not evidence for

Five analyses agreeing is only evidence in proportion to their independence, and ours are not independent. We set out the dependency structure explicitly rather than let the word "convergent" do work it cannot do.

Shared inputs and shared failure modes. Four of the five draw on the same composited NO_2 cube, over the same AOI, and three of the five subtract the same 25–60 km background ring; four of the five score against the same CREA benchmark; all five are judged by a single implementation of the robust-inference stack. Consequently:

- If the background ring contains the cluster's own advected plume (Section 5.3i), that error

propagates identically into Sections 5.2, 5.7 and 5.8, and their agreement is guaranteed a priori rather than informative.

- If part of the 93–95% common mode is retrieval-borne (Section 5.3ii), the same applies.
- If the CREA trader survey is a poor measure of physical output, every analysis that scores against it inherits that, and their agreement again tells us nothing new.
- A bug or a mis-specification in the shared inference implementation would likewise be invisible to all five.

Failure modes that are *not* shared. Against these specific alternatives the convergence does carry weight:

- **Section 5.9 (flux divergence)** uses raw Level-2 swaths, not the composited cube; no background ring; no deseasonalisation; and no benchmark at all — it is scored entirely on the internal source-versus-field-noise contrast. A ring artefact, a compositing artefact, a deseasonalisation choice or a benchmark defect cannot produce its result. It has its own, separate weakness (N=4; Section 5.9).
- **Section 5.6 (thermal proxy)** uses a different instrument, a different physical observable and a different failure mechanism entirely; its null is diagnosed positively (a seasonal crop-burning climatology), not merely observed.
- **Section 5.1 (scale sweep)** varies the AOI extent and grid resolution that Sections 5.2–5.8 hold fixed, so it rules out one specific shared parameter choice.
- **Section 5.4 (ring sweep)** varies the one remaining shared geometric parameter, the background-ring radii, across seven complete annuli out to 150 km. It does not clear them: it shows the cluster-local share *is* ring-dependent and does not converge, so this shared choice is now a known, quantified source of common error rather than an unexamined one.
- **Section 5.5 (scale contrast)** varies the shared *target*, scoring the same features against national steel activity. It leaves the shared publisher (CREA/WIND) untouched, because the independent-publisher arm returned NO-GO.
- Within the cube-based analyses, the change bases differ (first difference vs deseasonalised residual) and the cadences differ (daily vs weekly), which rules out the narrower class of artefacts specific to one of those choices.

The defensible summary is therefore narrower than "five independent confirmations". It is: *the cluster signal is not recoverable by any of the five routes we tried, and the two routes with genuinely disjoint failure modes (flux divergence and the thermal proxy) agree with the three that share inputs.* Whether the shared-input result reflects atmospheric coherence, ring self-subtraction or retrieval smoothness is *not* settled by this study — the ring sweep of Section 5.4 was designed to settle it, was re-run on a deliberately widened cube, and returned domain-limited both times. Section 7 names what remains.

7 Limitations

We separate limitations we can bound from controls we did not run. The second list is the more important one.

7.1 Bounded limitations

- **Single cluster.** These findings concern isolating *one* cluster (Tangshan, ∼13% of national steel) from a dense airshed, not the national or provincial NO_2–industry link, which the literature supports (Morris and Zhang, 2019; Li and Zheng, 2023) and which we do not contest.
- **Flux averaging depth.** N=4 of an intended 10–12, portal-throttled; see Section 5.9 for what this does and does not permit us to conclude.
- **Curtailment calendar.** The event ground truth comprises 21 sourced structural events (public holidays and heating-season restriction windows); a finer campaign-level calendar was not assembled. No event-level dates were imputed.
- **Benchmark provenance.** The CREA rate is a commercial trader survey. Section 5.6 attempted to arbitrate feature versus target with an external physical proxy and failed for instrument reasons. What we can now bound is the target's *reliability*: it tracks a separately surveyed 247-mill national panel at $r = 0.76$ in weekly changes, so measurement error in the benchmark caps the attainable correlation near 0.87 and cannot account for the near-null (Section 5.6). Its *validity* remains open, because both panels share a publisher; only an independent publisher closes that, and the NBS arm returned NO-GO.
- **Processor-version inhomogeneity.** Declared in Section 2. It invalidates secular-trend statements, which we have withdrawn; correlation and variance-share results between two series on the same cube are largely insulated from it.

7.2 Discriminating controls: three run, five outstanding, four specified

Each of the following would discriminate between our interpretation and a live alternative. C1 and C2 have since been run and are reported in Sections 5.4 and 5.5; neither corroborated the spatial reading, and both are summarised here with what they leave open. The benchmark-reliability bound of Section 5.6 has also been run and does discharge one alternative — that the target is too noisy to correlate with anything. C3–C7 have not been performed, and we do not claim the corresponding alternatives are excluded.

We add one control the earlier list did not contain, and place it first because it is the one whose absence most weakens the paper.

C0. Pipeline positive control — *not run, and the most consequential omission.* Every result in this paper is a null, which means a defective feature path and a genuinely absent steel signal are observationally identical here. Nothing in the study currently distinguishes them, and Section 5.5's *pipeline-suspect* verdict is the direct consequence. The discriminating test is cheap and needs no new data: run the identical footprint-minus-ring pipeline against NO_2 perturbations over this same AOI that are large, externally documented and not in dispute — the February 2020 control period, the Spring Festival cycle, the 2022 Winter Olympics controls. If the pipeline recovers them, the null in this paper is a statement about the steel signal. If it does not, the paper's central claim is not interpretable. We state plainly that we have not run it, and that until it is run the results here should be read as conditional on the pipeline being sound.

C1. Background-ring radius sensitivity — *run*, Section 5.4. Verdict: domain-limited. Run twice. The first pass, on the 0.25° AOI, returned *inconclusive* and exposed two defects in its own design: its far rings were corner wedges rather than annuli, and its pre-registered rule thresholded the share's *level* where it should have thresholded its *rise*. We recorded that defect rather than patching it — swapping the rule after seeing the data would have replaced a pre-registered verdict with a chosen one — and instead re-acquired the record over a 1.25° buffer,

pre-registering the rise rule and a plateau criterion before the new cube existed. The second pass confirms ring-dependence on complete annuli (weekly deseason share 0.007 → 0.061, CI excluding zero from 40–80 km) but still returns no quorum and no plateau. What remains is now a physical limit rather than a computational one: at 90–150 km the ring spans open sea and the Tianjin plume, so a wider cube would confound the test rather than settle it.

C2. Provincial and national benchmark arm — *run*, Section 5.5. Verdict: pipeline-suspect. Widening the target does not strengthen the coupling; the national pairings are near-null too and the paired contrast is negative throughout. What remains: the independent-publisher arm (NBS Hebei/national) was gated on a live probe that returned NO-GO — every NBS data endpoint refuses programmatic access while static assets on the same host serve normally — so the shared-publisher provenance gap is still open, and the decisive form of the test needs a province- or nation-scale NO_2 cube rather than a Tangshan window scored against a national target.

C3. Realistic-amplitude injection into real Level-2 fields. Inject a source of Tangshan's plausible emission magnitude into the four real overpasses and measure the recovery rate, to convert Section 5.9's non-detection into a power statement (Section 5.9). The existing synthetic control is a correctness check on an idealised field, not a power calculation.

C4. Wind-rotated composite. Align each overpass's source signature by wind direction before averaging, to test whether the faster-than-$1/\sqrt{N}$ noise decline is cancelling the source term rather than the background (Section 5.9).

C5. Retrieval-artefact decomposition. Propagate the per-overpass stratospheric column and air-mass-factor fields through the same compositing to estimate how much of the 93–95% common mode is retrieval-borne rather than atmospheric (Section 5.3ii).

C6. Multiplicative seasonal-lifetime interaction. The tropospheric NO_2 lifetime is roughly two to three times longer in winter than in summer, so the column-to-emission relationship is multiplicative and seasonal. Blast-furnace curtailment is *also* seasonal, concentrated in the heating season — precisely the season of maximum column sensitivity per unit emission. Our deseasonalisation is additive (an annual harmonic removed from the level, and a linear meteorological control), and an additive deseasonalisation does not remove an interaction between two seasonal cycles. The direction of the resulting bias is not obvious: a seasonally-varying gain could either amplify the curtailment signal or, if the harmonic fit absorbs the seasonal component of the response along with the seasonal component of the emissions, suppress exactly the signal we are looking for at the time of year it should be strongest. Testing this requires a multiplicative (log-space) or explicitly lifetime-weighted formulation, which we did not run. This is not addressed anywhere in the analyses above, and it is a plausible route by which a real signal could have been cancelled — the only one we have identified that could *reverse* the null rather than merely qualify it. The decisive step in any such test is a synthetic control in the opposite direction to the usual one: inject a known seasonal *multiplicative* gain and confirm that the additive estimator loses it. Without that step a negative result from the multiplicative arm would be uninterpretable.

C7. Processor-version harmonisation. Re-acquire with per-overpass processor version recorded, and apply the published inter-version corrections, before making any claim about secular change (Section 2).

8 What could reopen the question

Within this dataset. The priority is C0. Until the pipeline is shown to recover a NO_2 perturbation it ought to recover, every other control refines the interpretation of a result whose validity

is unestablished, and Section 5.5's *pipeline-suspect* verdict stands undischarged. It is also the cheapest of the eight: it needs no acquisition and no new estimator.

C1 and C2 have been run and did not corroborate the spatial reading (Sections 5.4, 5.5). What each leaves open is an independent-publisher benchmark, so the CREA/WIND provenance gap closes — noting that the reliability half of that concern is now bounded (Section 5.6) and only the validity half remains. The wider cube that C1 asked for has since been acquired and swept, and it closed the computational question without closing the physical one — the share does not plateau by 150 km, and past that radius the ring is no longer the same air mass. Pushing the ring further is not a route we recommend. Of the remainder, C3 and C4 are what would make Section 5.9 a powered result, and C6 is the one physical mechanism we identify that could plausibly have cancelled a real signal outright.

Unlikely to help. Additional TROPOMI overpasses under the current processing, finer de-seasonalisation of the additive kind, de-weathering (bounded to ∼0.4% of change-variance by Section 5.2), a tighter AOI (Section 5.1), or a different statistical estimator. Each was tried; none addresses a spatial common mode.

Potentially decisive, new instrument. A geostationary NO_2 sensor with sharper source–background contrast — notably GEMS, the geostationary spectrometer over East Asia (∼8 scans per day) — combined with plume-resolving attribution (Park et al., 2025; Beirle et al., 2023). Its gain is *diurnal frequency* rather than spatial resolution (∼10 km at Tangshan's off-nadir geometry, cluster- not plant-scale), so it targets a different lever (separating flat baseload industry from cyclical traffic through the diurnal cycle) rather than the spatial common mode directly; the expected value, given Section 6, is therefore uncertain. A free high-temperature industrial thermal product would independently reopen Section 5.6. Neither is available to a free-data study today.

9 Conclusion

A public, free-data satellite-NO_2 analysis, built at fixed parameters and scored blind to its benchmark, does not isolate the Tangshan steel-sector activity signal from the North China Plain regional background, and shows no demonstrable skill over the operational benchmark. Five analyses along five different recovery routes return the same answer. That much is robust: it does not depend on the background ring, the choice of target, or any single change basis.

Stated as the exclusion it is rather than as a failure to reject: at an effective sample size of ≈ 289 weeks, these data rule out any weekly linear coupling above $|r| = 0.17$ — under 2.8% of shared variance — between a public-data Tangshan NO_2 feature and the cluster's blast-furnace operating rate (Table 3). That bound is the falsifiable content of this paper, and it is not a statement about the benchmark's quality: the CREA Tangshan rate tracks a separately surveyed 247-mill national panel at $r = 0.76$ in weekly changes (Section 5.6), so its own measurement error caps the attainable correlation near 0.87, five times the bound we exclude.

What we cannot do is name the cause with confidence. The most economical account — that the binding constraint is spatial, the cluster and its background being inseparable at TROPOMI's resolution — had two cheap, discriminating predictions, and we tested both. Neither came back in its favour: the cluster-local variance share turns out to depend on where the background ring is drawn (Section 5.4), and the same feature tracks national steel activity no better than it tracks Tangshan's own rate (Section 5.5). The spatial reading survives as the leading hypothesis, not as an established conclusion, and the alternatives it was meant to displace — ring self-subtraction and a retrieval-borne common mode — remain live.

We report it this way deliberately. A null result is only worth publishing if its own weak

points are stated, and the two experiments most likely to embarrass this paper's interpretation are the two we ran. Section 7.2 lists the five that remain, and Section 8 names the instrument and the data that would settle the question. Well-characterised null results are how a measurement discipline learns where its instruments stop; a null whose own weak points are left unstated is not one of them.

A Complete scale-contrast family

Table 11: All 24 feature × target × sub-period pairings over which the Benjamini–Hochberg correction of Section 5.5 is computed. "nom." marks a pairing significant at $\alpha = 0.05$ on the block-permutation test *before* correction. Three pairings are nominally significant; none survives the correction, and none of them is a Tangshan pairing. Monthly (ME) rows carry $n \leq 89$ and are correspondingly weak.

Feature	Target	Period	cad.	n	r	p (perm)	nom.	BH p
intensity	Tangshan BF (cluster)	2019-2021	W	152	+0.068	0.439	–	0.554
intensity	Tangshan BF (cluster)	2022-2026	W	226	+0.057	0.410	–	0.554
intensity	Tangshan BF (cluster)	full	W	379	+0.062	0.291	–	0.466
intensity	national BF rate	2019-2021	W	152	−0.142	0.144	–	0.383
intensity	national BF rate	2022-2026	W	229	+0.024	0.764	–	0.864
intensity	national BF rate	full	W	382	−0.094	0.105	–	0.366
intensity	national crude steel	2019-2021	ME	36	−0.392	0.036	yes	0.366
intensity	national crude steel	2022-2026	ME	53	+0.065	0.657	–	0.789
intensity	national crude steel	full	ME	89	−0.222	0.062	–	0.366
intensity	national pig iron	2019-2021	ME	36	−0.365	0.059	–	0.366
intensity	national pig iron	2022-2026	ME	53	+0.188	0.182	–	0.401
intensity	national pig iron	full	ME	89	−0.154	0.227	–	0.419
residual	Tangshan BF (cluster)	2019-2021	W	145	+0.098	0.291	–	0.466
residual	Tangshan BF (cluster)	2022-2026	W	218	−0.016	0.828	–	0.864
residual	Tangshan BF (cluster)	full	W	364	−0.014	0.825	–	0.864
residual	national BF rate	2019-2021	W	145	−0.125	0.200	–	0.401
residual	national BF rate	2022-2026	W	221	−0.005	0.943	–	0.943
residual	national BF rate	full	W	367	−0.090	0.107	–	0.366
residual	national crude steel	2019-2021	ME	36	−0.343	0.103	–	0.366
residual	national crude steel	2022-2026	ME	53	+0.200	0.186	–	0.401
residual	national crude steel	full	ME	89	−0.120	0.349	–	0.523
residual	national pig iron	2019-2021	ME	36	−0.328	0.122	–	0.366
residual	national pig iron	2022-2026	ME	53	+0.306	0.037	yes	0.366
residual	national pig iron	full	ME	89	−0.110	0.415	–	0.554

Data and code availability

All inputs are public and free: TROPOMI tropospheric NO_2 from the Copernicus Data Space Ecosystem (`SENTINEL_5P_L2` via openEO, and the OData/zipper route for Level-2 swaths), ERA5 reanalysis from the Copernicus Climate Data Store, NASA active-fire detections, and the CREA blast-furnace operating-rate benchmark (`https://energyandcleanair.org/`; the pinned CSV export is cited in CREA, 2026). Analysis code and the derived result tables quoted here are in the project repository at `https://github.com/abetatos/NOXUS`, at commit `b4c1dd7`; the three intermediate NO_2 series marked below are regenerable but not redistributed.

Artefact-to-result map. So that no number in this paper is unanchored, every result section names the file it is computed from. Paths are relative to `data/derived/` unless marked otherwise.

Section	Artefact
5.1 scale sweep	`scale_sensitivity.csv`
5.2 variance	`variance_decomposition.csv`
5.4 ring sweep	`wide_ring_sensitivity.csv`; `ring_sensitivity.csv` (superseded first pass)
5.5 + App. A	`scale_contrast.csv`
5.6 thermal proxy	`viirs_triangulation.csv`
5.6 benchmark reliability	`benchmark_reliability.json`
5.7 regime stability	`regime_study.csv`; `regime_event_study.csv`
5.8 forecasting skill	`nowcast_probe.txt`; `catalyst_results.json`
5.9 flux divergence	`fluxdiv/`
4, Table 3	`no2/steel_activity_index.parquet`; `no2/steel_footprint_signal.parquet`; `no2/steel_intensity_decomposition.parquet` (∗)

(∗) These three series are *not* committed. They are intermediate products of the downloaded NO_2 record, and the project's data policy publishes derived *statistics* rather than redistributing reprocessed Copernicus holdings. They are regenerated from the public archive by `noxus fetch`, `noxus grid`, `noxus deseason` and `noxus attribute` at the committed configuration; the acquisition step needs free CDSE credentials. Every other artefact in the table is committed. We state this rather than let "in the repository" cover a file that is not there: a reader checking Table 3 must run four commands first, and should know that before starting.

Table 3 and the benchmark-reliability result are then reproduced by `uv run python analysis/benchmark_reliability.py`.

We flag one stale artefact rather than leave a reader to trip over it: `steel_validation_summary.txt` records an earlier run at a 5 km facility footprint radius and reports $r = 0.072$ at $n = 358$. The footprint of record in this paper is 15 km (Section 2), giving $r = 0.078$ at $n = 364$; the older file is retained for history and is not the source of any number quoted here.

On what "fixed-parameter" can be verified against. Section 3 claims that every parameter was fixed before the benchmark was read, and declines to call this preregistration. The verifiable substitute is the repository's own history: each compiled-in default lives in `noxus/config/run.py` and its introducing commit carries an author timestamp that precedes the commit introducing the corresponding validation call. That is a weaker guarantee than a registry entry — a determined author can rewrite history — but it is checkable by a third party, which a bare assertion is not, and we prefer to point at it than to ask for trust.

A citable archived snapshot with a DOI has not yet been minted; until one is, the commit above is the reproducibility anchor, and we note that a bare repository URL is a weaker guarantee than a versioned archive. All stochastic procedures use fixed random seeds with 5,000 bootstrap/permutation draws. The single candidate dataset requiring a licence (VIIRS Nightfire) was deliberately excluded to preserve reproducibility.

Declarations

Funding. None. **Competing interests.** None declared. **Author contributions.** Sole author.

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